



SCHOOL OF
**COMPUTING &
DATA SCIENCE**
The University of Hong Kong

COMP7507

Visualization and Visual Analytics

2026-27 First Semester

Course Information

Teaching Team

Instructor	Dr. Loretta YK Choi ykchoi@cs.hku.hk
Teaching Assistants	Mr. Li Jiuru u3011183@connect.hku.hk
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Please make appointments for consultation by email beforehand.

Course Format

Lectures

- f2f
- recordings will be available (on Moodle)

Hands-on exercises

- lab worksheets + submissions

Reference

- Matthew Ward, Georges Grinstein, and Daniel Keim, "Interactive Data Visualization: Foundations, Techniques, and Applications", A. K. Peters, Ltd.

Course Learning Outcomes

- Be able to understand the **key concepts** about visualization techniques and visual analytics.
- Be able to identify and evaluate the **key issues** in the application of data visualization techniques.
- Be able to **use appropriate visualization tools effectively** for data analysis.

Schedule (Tentative)

* via zoom

Session	Date	Topic	Hands-on modules	Project
1	Sep 9	Introduction	L0: Tableau installation	
2	Sep 19	Perception & Color	L1: Exploring the Dataset & Tableau Basics	Grouping Fixed
3	Oct 3	Temporal & Multivariate Data	L2: Vibe Coding: plotly and dash	
4	Oct 10	Interactions	L3: Building an Interactive Dashboard in Tableau	Proposal Due
5	Oct 17	Trees	L4: Vibe coding: design a web-based dashboard	In-class discussions
	Oct 24			In-class discussions *
6	Oct 31	Networks	L5: Network Visualization with Gephi	
7	Nov 7	Text & Document		In-class discussions
8	Nov 14	Geospatial & Volume Data		In-class discussions
9	Nov 21	Demo presentation		
10	Nov 28	Demo presentation		Final work, Critique

Prerequisite

Programming?

- Basic programming so you can write your own code/scripts for the project.
- Will cover plotly for data visualization in the course.

Mathematics?

- Basic mathematics or statistics for data aggregation.

Assessment

Final Examination [50%]

Continuous Assessment [50%]

Hands-on modules	5%	Individual
Group Project (45%)		
Proposal	5%	Group *
Demo Presentation	10%	Group *
Final Report + Work Product	20%	Group & Individual *
Critique	5%	Individual
In-class Discussion Participation	5%	Individual
* up to 15% of project marks moderated by peer review among group members		

Course Project

- Work in groups of 5
- Graded individually
- The assessment will be based mainly on the followings:
 - how well you can apply proper visualization design rationales and techniques
 - how well you can perform visual analysis (e.g., hypothesis generation, verification, knowledge discovery)
- Read the project requirements carefully at each stage

Late Submission Policy

- All checkpoint and project submissions must be handed in by the deadlines.
- Late Policy: 50% deduction if submitting within 2 days after the deadline; no mark thereafter.

Use of Gen AI

- Encouraged but please follow the HKU policies and make sure academic integrity is maintained.
- <https://tl.hku.hk/genai/>
- <https://talic.hku.hk/innovation/guidance-for-ethical-use-of-ai-single-page/>

Plagiarism

You are allowed and encouraged to discuss your work with other students and the TAs. However, your submission must be your own original work.

- As defined in the University's Regulations Governing Conduct at Examinations, plagiarism is the **unacknowledged use, as one's own, of work of another person, whether or not such work has been published**. Or put it simply, plagiarism is **copying the work of another person without proper acknowledgement**. In case of queries on plagiarism, students are strongly advised to refer to ["What is Plagiarism?"](#).
- **First Attempt:** Students who admit committing plagiarism for the first time shall be warned in writing and receive a **zero mark for the component concerned**. For those who do not confess, the case would be referred to the Programme Director for consideration.
- **Subsequent Attempt:** If students commit plagiarism more than once during the course of studies, the case shall be referred to the Programme Director for consideration. The Programme Director will investigate the case and consider referring it to the University Disciplinary Committee, which may impose any of the following penalties: **a published reprimand, suspension of study for a period of time, fine, or expulsion from the University**.

Communication Channel

- Moodle Discussion Forum (most preferred channel)
- Email
- Face-to-face Meeting
 - consultation hours
 - make appointment by email for specific time slots